

MH1812 Discrete Mathematics

– Continuous Assessment 2

NTU, AY19/20 S1

31st Oct 2019, 10:30AM - 11:30AM @ Exam Hall C, LT2A

Name:

Email:

Tutorial Group:

*Note: There are **6 pages** in total. The **first 5** questions are compulsory. Question 6 is bonus, you get the points if you answer it correctly, you don't lose any points if you leave it or answer wrongly. Write on the back of paper if there is not enough space. Leave the tutorial group if you can't remember.*

Question 1. (16 points)

Given the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$, with $a_1 = 5$ and $a_0 = 2$, find the explicit formula of a_n .

Solution: The characteristic equation of the above relation is:

$$x^2 - 5x + 6 = (x - 2)(x - 3) = 0$$

which has two solutions $s_1 = 2$ and $s_2 = 3$, hence

$$a_n = u2^n + v3^n.$$

substitute $a_1 = 5$ and $a_0 = 2$, we obtain

$$a_1 = u2^1 + v3^1 = 2u + 3v = 5 \text{ and } a_2 = u2^0 + v3^0 = u + v = 2$$

solving above gives $u = 1$ and $v = 1$, hence $a_n = 2^n + 3^n$. □

Question 2. (20 points)

In all positive integers no larger than 2019, i.e., $\{1, 2, 3, \dots, 2019\}$,

1. How many numbers are multiple of 3 ?
2. How many numbers are multiple of 5 ?
3. How many numbers are multiple of 15 ?
4. How many numbers are multiple of 3 or 5 ?

Solution:

1. $|S_3| = \lfloor \frac{2019}{3} \rfloor = \lfloor 673 \rfloor = 673$
2. $|S_5| = \lfloor \frac{2019}{5} \rfloor = \lfloor 403.8 \rfloor = 403$
3. $|S_{15}| = \lfloor \frac{2019}{15} \rfloor = \lfloor 134.6 \rfloor = 134$
4. $|S_3 \cup S_5| = |S_3| + |S_5| - |S_3 \cap S_5| = |S_{15}| = 673 + 403 - 134 = 942$

□

Question 3. (26 points)

Defined on set $A = \{a, b, c, d\}$, the binary relation $R = \{(a, b), (b, a), (b, c), (c, d)\}$.

1. Is R reflexive, symmetric, transitive, anti-symmetric ? Justify your answer by giving a counter-example when it does not hold the property.
 2. Find R^2 , which is defined to be $R \circ R$.
 3. Find the transitive closure R^t .
1. R is NOT reflexive, $a \in A$ but $(a, a) \notin R$.
 R is NOT symmetric, $b, c \in A$ and $(b, c) \in R$ but $(c, b) \notin R$.
 R is NOT transitive, $b, c, d \in A$ and $(b, c), (c, d) \in R$ but $(b, d) \notin R$.
 R is NOT anti-symmetric, $a, b \in A$ and $(a, b), (b, a) \in R$ but $a \neq b$.
 2. $R^2 = \{(a, a), (a, c), (b, b), (b, d)\}$
 3. $R^t = \{(a, a), (a, b), (a, c), (a, d), (b, a), (b, b), (b, c), (b, d), (c, d)\}$

Question 4. (20 points)

Let set $A = \{a, b\}$ and $B = \{b, c\}$,

1. Give $P(A)$, $P(B)$, and $P(A) - P(B)$ by listing their elements.
2. Give $A - B$ and $P(A - B)$ by listing their elements.
3. Does $P(A) - P(B) = P(A - B)$?
4. List the elements of $(A - B) \cup (B - A)$ and $(A \cup B) - (A \cap B)$. Are they equal?

Solution:

1. $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
 $P(B) = \{\emptyset, \{b\}, \{c\}, \{b, c\}\}$
 $P(A) - P(B) = \{\{a\}, \{a, b\}\}$
2. $A - B = \{a\}$
 $P(A - B) = \{\emptyset, \{a\}\}$
3. NO, $\{a, b\} \in P(A) - P(B)$ but not in $P(A - B)$.
4. $A - B = \{a\}$ and $B - A = \{c\}$, so $(A - B) \cup (B - A) = \{a, c\}$
 $A \cup B = \{a, b, c\}$, and $A \cap B = \{b\}$, so $(A \cup B) - (A \cap B) = \{a, c\} = (A - B) \cup (B - A)$.

□

Question 5. (18 points)

1. How many permutations are there of the 3 letters “abc”?
2. How many permutations are there of the 4 letters “abcd”?
3. How many permutations are there of the 4 letters “abcc”?

Solution:

1. $3! = 3 \cdot 2 \cdot 1 = 6$
2. $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$
3. $4!/(2!) = (4 \cdot 3 \cdot 2 \cdot 1)/(2 \cdot 1) = 12.$

□

Bonus Question 6 (10 points)

Defined on a set of size 4, how many equivalence relations are there ?

Solution: Each equivalence relation is a partition of the set, so the question is about counting how many possible partition of the set. This can be done by dividing them by the size of partitions.

$$1 + 1 + 1 + 1: 1$$

$$2 + 1 + 1: \binom{4}{2} = 6$$

$$2 + 2: \binom{4}{2}/2 = 3$$

$$3 + 1: \binom{4}{3} = 4$$

$$4: 1$$

Hence in total $1 + 6 + 3 + 4 + 1 = 15$.

□